

Food-Delivery Apps and Small-World Networks: How Uber Eats or Wolt Change Local Business Connections

a) Importance and relevance of the problem

Apps like Uber Eats, Wolt, and Bolt Food changed how people order food and how restaurants run their business. They connect thousands of restaurants, couriers, and customers through one digital system, like a big living network. These links help new ideas or food trends move super-fast but they also make restaurants depend on the app too much. I want to understand if these strong and close connections (small-world structure) help small restaurants grow or just make them more stuck in the system.

b) What we already know

From the small-world idea (Watts & Strogatz 1998), when a network has high clustering and short paths, info spreads really fast. In economic-complexity theory (Hidalgo & Hausmann 2009), countries or regions grow when they combine what they already know to make something more advanced. If we mix these two ideas, delivery networks could help restaurants learn faster — like digital menus, ads, or better logistics which can make them more “complex” and modern.

c) Niche and research question

Most people study how customers use these apps, but not much about how the *network structure* looks inside.

Research question: *Do small-world connections between restaurants and couriers make restaurants adopt new digital tools faster and grow their business easier?*

It could be tested by building a small local delivery network, checking who connects with who, and seeing if more-connected ones improve faster.

d) Stakeholders and potential benefits

This topic matters for restaurants, couriers, platform companies, and city policy people. Restaurants can learn how to use apps smarter; platforms can make their systems fairer; and cities can understand how digital work changes local economies. Everyone gets an idea of how “being connected” helps or hurts small business life.

e) References

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Ismet Ogulcan TOGAN

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